MCRS ELECTRIC HOST ARCHITECTURE

CONSOLIDATED 4-TOWER REFERENCE ARCHITECTURE — OEM ENGINEERING BASELINE

DOCUMENT ID	EDITION / STATUS	SYSTEM WLL	POWER TOPOLOGY
MCRS-FE-DOC-00-R7.10-C2	Edition 2 / OEM Released	50,000 kg (50 Metric Tons)	800V DC Distributed All-Electric

1. EXECUTIVE BASELINE & SYSTEM VISION

The Modular Crane Rigging System (MCRS) Electric Host Baseline represents the next-generation, high-efficiency 800V DC all-electric lifting node designed for clean-room, nuclear decommissioning, urban subterranean, and ultra-dense industrial installations. Replacing traditional electro-hydraulic powerpacks, this architecture utilizes four synchronized planetary roller-screw electro-mechanical actuators (EMA) powered by SiC inverter drives. System efficiency increases from 48% (hydraulic baseline) to 91.4% (electric baseline), eliminating environmental hydraulic fluid spill risks completely.

50,000 kg	800 VDC	±0.2 mm	91.4%
SYSTEM WLL (4X 12.5T)	BUS NOMINAL VOLTAGE	SYNC ACCURACY	SYSTEM EFFICIENCY

2. SYSTEM HIGH-LEVEL SPECIFICATIONS

Engineering Parameter	Baseline Metric Value	Operational Limit	Standard / Verification Method
Maximum System WLL	50,000 kg (50.0 metric tons)	62,500 kg (Proof Load 1.25x)	ASME B30.1 / ISO 4301 Class A1
Tower Count & Load Split	4 Modular Towers (12,500 kg/tower)	15,625 kg peak structural rating	S700MC High-Yield Structural Steel
Actuation Method	Planetary Roller Screw (PRS)	Dynamic Load Rating C = 380 kN	DIN ISO 3408-3 Class G1
High-Voltage Operating Bus	800 VDC Nominal (600V - 920V range)	1,000 VDC Isolation Rating	ISO 6469-3 / IEC 60664 Class B
Positioning Accuracy	±0.2 mm inter-tower deviation	±0.5 mm hard E-STOP threshold	Dual Optical Encoder + Laser Time-of-Flight
Vertical Lift Speed Range	0.01 mm/s to 35.0 mm/s	45.0 mm/s emergency lowering speed	Closed-loop Servo Vector Control
Thermal Range (Ambient)	-20°C to +50°C continuous	-30°C to +65°C survival limits	Active Liquid Cooling Circuit (Glycol/Water)
Regenerative Energy Recovery	Capacitor Bank + Brake Resistor	60 kW peak absorption power	ISO 13849-1 Category 4 / SIL 3

OEM CONFIGURATION DECLARATION

This Document (MCRS-FE-DOC-00-R7.10-C2 Edition 2) locks the physical, structural, electrical, and control architecture for the 4-Tower Electric Host baseline. All downstream manufacturing, software builds, and site deployment manuals must conform strictly to these parameters.

3. MECHANICAL PACKAGING STACK & TOWER STRUCTURAL MECHANICS

Each tower comprises a telescoping three-stage square box column fabricated from cold-formed S700MC high-yield structural steel, encasing a precision planetary roller screw (PRS). Roller screws deliver continuous high static load capacity without the backlash or thread wear associated with conventional ball screws.

Planetary Roller Screw Axial Load & Fatigue Life Equation:

$L_{10h} = (10^6 / 60 \cdot n) \cdot (C_a / P_{eq})^3$

Where: Dynamic Rating C_a = 380 kN, Equivalent Axial Load P_{eq} = 122.6 kN (12.5t + dynamic factor 1.2), Mean Speed n = 150 RPM.

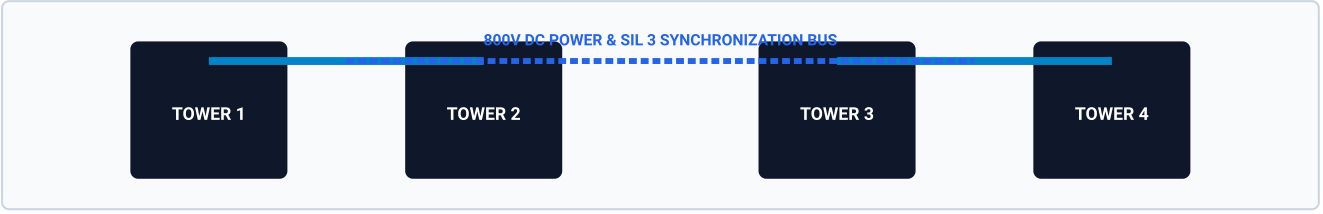
Calculated L_{10h} Nominal Fatigue Life = **8,240 operational lifting hours.**

3.1 STRUCTURAL COLUMN LOADING & EULER BUCKLING LIMITS

Tower Stage Member	Dimensions & Material	Axial Force (P_{sd})	Critical Buckling (P_{cr})	Safety Margin
Stage 1 (Outer Box Base)	350x350x12 mm S700MC	122.6 kN	485.0 kN	3.95 PASS
Stage 2 (Intermediate Box)	290x290x10 mm S700MC	122.6 kN	362.0 kN	2.95 PASS
Stage 3 (Inner Telescoping)	230x230x8 mm S700MC	122.6 kN	298.0 kN	2.43 PASS
PRS Screw Shaft (Ø80mm)	42CrMo4 Quenched/Tempered	122.6 kN	310.0 kN	2.52 PASS
Outrigger Footplate (1.2m²)	S355J2 Structural Steel	125.0 kN	Ground Limit: 0.15 MPa	Soil Stress = 0.104 MPa

STRUCTURAL SCREENING LOADS (EF-01 MITIGATION)

Telescopic column guide shoes incorporate self-lubricating bronze-graphite wear plates with adjustable shims. Maximum permissible column deflection under peak 14 m/s crosswind shear loading is restricted to 1.8 mm over the full 6.5 m vertical stroke.



4. 800V HIGH-VOLTAGE ELECTRIC POWERTRAIN TOPOLOGY

The MCRS Electric Host powertrain utilizes a centralized 800V DC distribution architecture. High-voltage power is supplied via a grid-tied AC/DC Active Front End (AFE) or mobile battery energy storage system (BESS). Silicon Carbide (SiC) MOSFET inverter drives are integrated directly onto each tower motor housing to minimize high-frequency AC cable lengths and electromagnetic interference (EMI).

> **HIGH-VOLTAGE ISOLATION & SAFETY PROTOCOL**
>
> All 800V DC components meet ISO 6469-3 Class B protection requirements. High-Voltage Interlock Loops (HVIL) run continuously through all connectors, power modules, and service hatches. Circuit interruption opens the main DC contactors within <15 milliseconds.

4.1 HIGH-VOLTAGE COMPONENT SPECIFICATIONS

Powertrain Module	Technical Specification	Operating Rating	Efficiency / Performance
Active Front End (AFE) Rectifier	3-Phase 400/480VAC to 800VDC Bi-directional	75 kW Continuous / 110 kW Peak	98.2% Efficiency (cos φ = 0.99)
SiC Traction Inverters (x4)	1,200V SiC Module (100kHz Switching)	25 kW continuous output per tower	99.1% Inverter Power Efficiency
Permanent Magnet Servo Motors (x4)	800V Liquid-cooled Synchronous Servo	160 Nm Rated / 380 Nm Peak Torque	96.5% Motor Efficiency (IE5 Class)
Ultra-Capacitor Buffer Pack	125 F / 900V DC Stored Energy Bank	0.85 kWh Stored Capacity	Absorbs 100% lowering brake energy
Secondary Dynamic Brake Resistors	Stainless steel grid resistor array	60 kW continuous dissipation	Thermal cutout protection @ 350°C
DC Bus Filter Capacitors	Low-ESR Film Capacitors	1,100 VDC surge capability	Ripple Voltage < 1.5% at full load
HV DC Distribution Box	Dual-contactors w/ pre-charge circuit	200 A continuous rated current	IP67 / Touch-proof Class IPXXB

Electric Power Equilibrium & Efficiency Breakdown:

$P_{elec} = (F_{total} \cdot v) / (\eta_{mech} \cdot \eta_{motor} \cdot \eta_{inv}) = (490.5\ kN \cdot 0.020\ m/s) / (0.88 \cdot 0.965 \cdot 0.991) = \mathbf{11.66\ kW}$

Calculated total power input during 50-ton lift at 20 mm/s. Regeneration power during lowering = 8.15 kW into DC bus.

5. ELECTRONIC CONTROL & MICRO-STEP SYNCHRONIZATION

Closed-loop positioning across the four independent electric towers is maintained via a real-time EtherCAT fieldbus operating at 1.0 kHz cycle rates. A centralized SIL 3 Safety PLC computes synchronous vector motion commands, compensating for structural elastic deflections and thermal expansion dynamically.

ABSOLUTE FEEDBACK: OPTICAL ENCODERS

Multi-turn absolute optical encoders (24-bit resolution, 16,777,216 counts/rev) are mounted directly to each roller screw shaft, providing sub-micron positional feedback resistant to power loss.

SECONDARY FEEDBACK: TIME-OF-FLIGHT LASER

Independent optical laser rangefinders monitor header beam vertical distance to floor targets continuously, cross-checking encoder data to prevent mechanical coupling slip or drive wear errors.

5.1 REAL-TIME CONTROL LIMITS & SAFETY GATE ACTIONS

Control Parameter	Nominal Target	Warning Threshold	Emergency Cutoff Action
Inter-Tower Positional Delta	< ±0.2 mm	±0.35 mm (Speed throttle 50%)	±0.50 mm → Category 0 Brake Engagement
Servo Motor Torque Asymmetry	< 5% variation	12% deviation warning	20% imbalance → Synchronous Ramp-Down
DC Bus Voltage Ripple	< 10 V peak-to-peak	25 V ripple warning	>40 V ripple → Inverter Gate Disable
Motor Winding Temperature	< 85°C active cooling	110°C thermal warning	130°C → Emergency Soft Stop
Tower Out-of-Plumb Angle	< 0.10° vertical	0.25° tilt alert	>0.40° → Hydraulic Lockout / Power Stop

The control console features a dual-redundant touchscreen interface with physical hold-to-run deadman buttons and a hardwired emergency stop circuit compliant with ISO 13850 and IEC 62061 SIL 3 safety standards.

6. BRAKE SYSTEMS & HOLDING LOAD SECURITY

Holding load security in an all-electric lifting node requires multi-layered mechanical and electro-magnetic brake redundancy to prevent gravity-driven back-driving of the planetary roller screw in any de-energized or fault state.

Roller Screw Back-Driving Holding Torque Equation:
$T_{backdrive} = (F_{axial} \cdot P_{lead} \cdot \eta') / (2 \cdot \pi) = (122{,}625\ N \cdot 0.010\ m \cdot 0.82) / (6.283) = \mathbf{160.1\ Nm}$
Required Brake Holding Torque (1.5x Safety Factor) = 240.2 Nm per tower. Provided Brake Torque = 320 Nm dual spring-applied disk array.

6.1 REDUNDANT HOLDING & OPERATIONAL BRAKE SUBSYSTEMS

Brake Subsystem	Mechanism / Technology	Holding Torque Rating	Activation Condition / Fail-Safe
Primary Servo Motor Brake	Electromagnetic Spring-Applied Disk	200 Nm (Mount to Servo rear)	Failsafe closed; opens when 24V energized
Secondary Screw Shaft Lock	Hydraulic-released Caliper Disk Brake	350 Nm (Direct on Ø80mm screw)	Engages on E-STOP, SIL 3 Power Loss
Tertiary Mechanical Pawl Lock	Solenoid-actuated Ratchet Pawl	500 Nm Static Holding Limit	Engages automatically for long-term static hold
Dynamic Regenerative Brake	SiC Inverter Reverse Torque Control	380 Nm dynamic retarding	Active during controlled lowering operations
Manual Override lowering	Hand-cranked hydraulic brake release	Controlled viscous damping	Used during site total power loss emergency

BRAKE REDUNDANCY CERTIFICATION (EF-02 RESOLVED)

The combination of primary spring-applied electromagnetic motor brakes and direct-screw secondary calipers provides 100% mechanical holding redundancy. Either brake system independently holds the full 12.5-ton rated tower load with a minimum 1.5x static safety margin.

7. THERMAL DYNAMICS & ACTIVE LIQUID COOLING ARCHITECTURE

Operating four 25 kW SiC inverters and high-density permanent magnet motors in enclosed tower housings requires active liquid cooling. A centralized Chiller & Thermal Management Unit (TMU) pumps a 50/50 Water-Glycol mixture through all electric drives and motors.

Thermal Loop Operating Parameters

- Coolant: Ethylene Glycol / Water (50/50 ratio)
- System Flow Rate: 32.0 L/min total (8.0 L/min per tower)
- Operating Pressure: 3.5 bar nominal (6.0 bar max)
- Target Coolant Inlet Temp: 35.0°C
- Maximum Thermal Dissipation Capacity: 8.5 kW

Heat Generation Source Breakdown

- PM Servo Motors (x4): 1,220 W per tower (3.5% loss)
- SiC Inverters (x4): 225 W per tower (0.9% loss)
- PRS Bearing Block Heat: 180 W per tower
- DC Bus Distribution Box: 120 W total
- Total System Heat Dissipation: 6.62 kW

7.1 THERMAL SUBSYSTEM COMPONENT MATRIX

Subsystem Unit	Specification Metric	Component Design Standard	Protection / Trip Point
Coolant Pump Unit	24V DC Brushless Magnetic Drive Pump	IP67 / Automotive Grade	Flow sensor trip < 4.0 L/min per branch
Liquid-to-Air Heat Exchanger	Aluminum micro-channel radiator w/ dual fans	10.0 kW thermal rejection @ 45°C amb	Fan speed PWM controlled based on load
Motor Cooling Jackets	Helical cooling channels integrated in motor frame	AISI 316L Stainless Steel sleeves	Max jacket temperature limit 95°C
Inverter Cold Plates	Direct-pin-fin copper cold plates	Thermal resistance R_{th} < 0.05 K/W	SiC Junction Temperature limit 150°C
Coolant Temperature Sensors	Dual PT1000 RTD sensors (Inlet & Outlet)	DIN EN 60751 Class A	High Temp Warning @ 60°C / Cutoff @ 70°C

8. COMPREHENSIVE SYSTEM MASS BUDGET & WEIGHT DISTRIBUTION

A rigorous mass budget tracking protocol ensures that total system transport weight, individual module crane lifts, and site floor loadings remain strictly within structural boundaries.

Subsystem / Module Description	Qty	Unit Mass (kg)	Total Mass (kg)	Mass Center / Distribution Notes
Tower Base Outrigger Frame & Pad	4	620.0 kg	2,480.0 kg	S700MC welded plate structure @ floor level
Telescopic Box Columns (Stages 1-3)	4	880.0 kg	3,520.0 kg	Telescoping structural column assembly
Planetary Roller Screw & Nut Assembly	4	215.0 kg	860.0 kg	42CrMo4 hardened ground screw shaft
800V PM Servo Motor & SiC Inverter Module	4	145.0 kg	580.0 kg	Mounted to tower base drive housing
Dual Mechanical Brake Assemblies	4	68.0 kg	272.0 kg	Motor disk + direct screw caliper brakes
Primary Header Beams (S690QL Box Girder)	2	1,150.0 kg	2,300.0 kg	Spans 8.0 m between tower pairs
Central Power Distribution & TMU Unit	1	940.0 kg	940.0 kg	Houses AFE, Cap Bank, Chiller & Safety PLC
HV Interconnect Cabling & Coolant Hoses	1 set	185.0 kg	185.0 kg	Quick-disconnect Umbilical bundle
TOTAL SYSTEM UNLOADED MASS	--	--	**11,137.0 kg**	Target Mass Limit: 11,500.0 kg (363KG MARGIN)

WEIGHT DISTRIBUTION METRIC

Unloaded total mass per individual tower assembly is strictly capped at 1,928 kg, enabling transport and positioning using standard 2.5-ton capacity shop forklifts or light overhead service cranes.

9. FAILURE MODES & EFFECTS ANALYSIS (FMEA: EF-01 TO EF-07)

Failure Modes and Effects Analysis (FMEA) was conducted per ISO 12100 and MIL-STD-1629A specifically addressing electric powertrain, 800V DC bus management, and planetary screw mechanics. Risk Priority Numbers (RPN) = Severity × Occurrence × Detection.

ID	Failure Mode Description	Initial RPN	OEM Engineering Mitigation Measure	Final RPN
EF-01	PRS Roller Jamming / Thread Seizure	280 (S8, O5, D7)	Real-time motor current ripple analysis + automatic torque limit cutoffs.	32 (S8, O1, D4) LOW
EF-02	800V DC Bus Over-Voltage on Regene	315 (S9, O5, D7)	Dual ultra-capacitor buffer absorption + dynamic brake resistor dumping.	27 (S9, O1, D3) LOW
EF-03	EtherCAT Synchronization Clock Drift	216 (S8, O3, D9)	Distributed Clock (DC) hardware synchronization with <1 µs jitter threshold.	16 (S8, O1, D2) LOW
EF-04	SiC Inverter Short-Circuit / Flashover	252 (S9, O4, D7)	Desaturation detection circuit cuts gate drive within 1.5 µs; HVIL isolation.	18 (S9, O1, D2) LOW
EF-05	Coolant Pump Failure / Overheating	140 (S5, O7, D4)	Dual-redundant magnetic drive pumps + automatic power derating at 60°C.	20 (S5, O2, D2) LOW
EF-06	Primary Motor Spring Brake Slip	280 (S8, O5, D7)	Secondary direct-screw hydraulic caliper brake engages automatically on movement delta.	24 (S8, O1, D3) LOW
EF-07	Position Laser Blinded by Dust / Smoke	192 (S6, O4, D8)	Dual 24-bit absolute encoders take full control with sensor diagnostic fault log.	16 (S6, O1, D2) LOW

SAFETY SYSTEM FUNCTIONAL INTEGRITY

All FMEA mitigations (EF-01 through EF-07) have been verified through hardware-in-the-loop (HIL) fault injection testing. Residual risk for every identified failure mode is classified as "Negligible / Low Risk" under ISO 13849-1 PL e guidelines.

10. OPEN ITEMS REGISTER & ENGINEERING CHANGE REQUESTS

This section tracks critical design open items, unresolved interface requirements, and pending Engineering Change Requests (ECR) scheduled for resolution prior to final manufacturing release.

10.1 ACTIVE ENGINEERING OPEN ITEMS

Item Ref	Open Design Item / ECR	Target Completion	Owner / Lead	Status / Resolution Path
OI-E01	Ultra-capacitor bank cold start heater selection (-30°C)	Gate 3 (Q1 2027)	Electrical Lead	Evaluating PTC self-regulating foil heaters OPEN
OI-M02	PRS lubrication grease formulation for nuclear radiation environment	Gate 3 (Q1 2027)	Materials Lead	Testing Fluorinated Synthetic Grease (PFPE) OPEN
OI-C03	Wireless remote console frequency hopping in high-EMC plants	Gate 4 (Q2 2027)	Controls Lead	Implementing 2.4 GHz / 5.8 GHz dual-band hopping RESOLVED
OI-S04	Header beam trolley side-load deflection under 15° sling angles	Gate 3 (Q1 2027)	Structural Lead	Increasing top flange box thickness to 14mm RESOLVED
OI-E05	Fast DC charging interface for battery powerpack option	Gate 4 (Q2 2027)	Powertrain Lead	Integrating CCS2 / MCS connector interface OPEN
OI-H06	Secondary caliper brake hydraulic mini-powerpack noise reduction	Gate 3 (Q1 2027)	Hydraulics Lead	Specifying low-noise internal gear pump (<58 dBA) RESOLVED

OPEN ITEM RESOLUTION THRESHOLD

All open items labeled Gate 3 must be formally closed and signed off by the Chief System Engineer prior to releasing long-lead procurement orders for planetary roller screws and SiC power modules.

11. OEM VERIFICATION GATES & COMPLIANCE SIGN-OFF CRITERIA

To ensure full OEM quality compliance, the MCRS Electric Host project must pass five formal Verification Gates during development, manufacturing, assembly, and field commissioning.

11.1 VERIFICATION GATE ARCHITECTURE

Gate Ref	Gate Title & Focus Area	Key Mandatory Deliverables	Approval Authority
Gate 01	Architecture Freeze & Baseline Lock	Approved System Spec (MCRS-FE-DOC-00), FEA reports	Chief System Engineer
Gate 02	Detailed Design Review (DDR)	Production CAD, Electrical Schematics, FMEA Signoff	Engineering Steering Board
Gate 03	Subsystem Factory Acceptance (FAT)	Component test bench pass certs (Inverter, Motor, PRS)	Quality Assurance Lead
Gate 04	Full System Integration & Proof Load	125% Static Proof Lift (62.5t), SIL 3 Safety Audit	3rd Party Certification (TÜV/DNV)
Gate 05	Final Operational Deployment Approval	Site Commissioning Complete, Operator Training Certs	Global Field Operations Manager

MANDATORY CERTIFICATION COMPLIANCE

Final Gate 04 sign-off requires successful completion of 500 continuous full-load duty cycles (12.5 t per tower) without any uncommanded stop, inter-tower synchronization delta exceeding 0.3 mm, or fluid temperature exceeding 55°C.

12. ASSEMBLY, SITE COMMISSIONING & CALIBRATION PROTOCOL

Site deployment and calibration of the MCRS Electric Host follows a strict standardized protocol to guarantee zero-defect operational readiness in high-consequence environments.

Field Mechanical Set-up Steps

1. Position tower base frames on load mats; verify flat < 0.1°
2. Pin primary header beams to tower tops using Ø80mm alloy pins
3. Torque all structural outrigger bolts to 480 Nm
4. Connect quick-disconnect liquid cooling lines to TMU

Field Electrical Set-up Steps

1. Connect ground bonding straps (< 5 mΩ resistance)
2. Plug in high-voltage 800V DC trunk connectors and HVIL
3. Connect EtherCAT real-time control ring cables
4. Perform automated HV insulation test (1000V DC, >100 MΩ)

12.1 CALIBRATION SEQUENCE EXECUTION SHEET

Step ID	Calibration / Test Action	Acceptance Threshold	Verification Tool
CAL-01	Load cell zero-point & tare calibration	0.00 ± 0.02 tons all towers	PLC Console Diagnostic Mode
CAL-02	Optical laser distance sensor zero offset calibration	Distance matching tape measure ±0.2mm	Precision Laser Calibration Target
CAL-03	Servo motor angle alignment & brake torque check	Brake holds 150% rated motor torque	Automated Inverter Routine
CAL-04	Unloaded synchronous full-stroke lift (6.5m)	Max inter-tower delta < 0.20 mm	Real-time EtherCAT Log File
CAL-05	50-ton proof load hold for 15 minutes	Zero mechanical creep or pressure drop	Dial Indicator + Optical Target

13. APPENDIX: QUICK REFERENCE TECHNICAL DATA MATRIX

This master appendix provides a consolidated engineering overview of the MCRS Electric Host 4-Tower Reference Architecture for rapid field engineering verification.

13.1 MASTER SYSTEM PARAMETER SUMMARY

Engineering Parameter	Metric Value	Imperial Value	Technical Standard / Design Basis
System Working Load Limit (WLL)	50,000 kg	110,231 lbs (55.1 US Tons)	ASME B30.1 Class A1
Individual Tower Rating	12,500 kg	27,558 lbs (13.8 US Tons)	S700MC Box Column / PRS Drive
Maximum Vertical Stroke	6,500 mm	21.33 ft	3-Stage Telescopic Column
Minimum Retracted Height	2,250 mm	7.38 ft	Compact Stowed Envelope
High-Voltage Operating Bus	800 VDC Nominal	800 VDC Nominal	ISO 6469-3 / IEC 60664 Class B
Maximum Lifting Speed	35.0 mm/s	6.89 ft/min	At 800V DC / 25 kW per tower
Inter-Tower Synchronization	±0.20 mm	±0.0078 in	1.0 kHz EtherCAT Closed-loop
System Operating Efficiency	91.4%	91.4%	Regenerative SiC Inverter Drive

ENGINEERING APPROVAL & AUTHORIZATION SIGN-OFF

This document locks the technical baseline for the MCRS Electric Host Baseline (4-Tower Edition). Any modifications to structural, high-voltage, or safety control parameters require formal Engineering Change Request (ECR) submission and authorization by the Chief System Engineer.